$$X_R^{+} \, B \, W_\gamma = 0$$

0th time derivative

$$\begin{bmatrix} X_R[1] \\ X_R[2] \\ \vdots \\ X_R[ORD] \end{bmatrix}^{+} \begin{bmatrix} I & & & \\ & I & & \\ & & \ddots & \\ & & & B[ORDP1] \end{bmatrix} \begin{bmatrix} W_\gamma^{(0)} \\ W_\gamma^{(1)} \\ \\ W_\gamma^{(ORD-1)} \end{bmatrix} \overset{!}{=} 0$$

(ORD-1) th time derivative

e.g. mass matrix

$$X_R^+[1]\ W_\gamma^{(0)} + X_R^+[2]\ W_\gamma^{(1)} + \cdots + X_R^+[ORD]\ B[ORDP1]\ W_\gamma^{(ORD-1)} \overset{!}{=} 0$$

$$X_r^+[ORD] \sum_{j=2}^{ORDP1} B[j]\,\tilde{\lambda}_r^{j-2}\ W_\gamma[1] + X_r^+[ORD] \sum_{j=3}^{ORDP1} B[j]\,\tilde{\lambda}_r^{j-3}\ W_\gamma[2] + X_r^+[ORD]\,B[ORDP1]\ W_\gamma[ORD]$$

$$= \sum_{j=1}^{ORD} \sum_{L=j+1}^{ORDP1} X_r^+[ORD]\ B[L]\ \lambda_r^{L-(j+1)}\ W_\gamma[j]$$

$$W_\gamma[j] = s^{j-1} W_\gamma[1] + \sum_{a=0}^{j-2} \left(Y[1]\,\Lambda^{j-2-a} F_\gamma + \mathcal{Z}_\gamma[j-1-a] \right) s^a$$

$$= \sum_{j=1}^{ORD} \sum_{L=j+1}^{ORDP1} X_r^+[ORD]\ B[L]\ \lambda_r^{L-(j+1)}\ s^{j-1}\ W_\gamma[1]$$

$$+ \sum_{j=\cancel{1}\,2}^{ORD} \sum_{L=j+1}^{ORDP1} \sum_{a=1}^{j-1} X_r^+[ORD]\ \lambda_r^{L-(j+1)}\ \underbrace{B[L]\,Y[1]\,\Lambda^{j-(a+1)}}\ s^{a-1}\ F_\gamma$$

$$+ \sum_{j=\cancel{1}\,2}^{ORD} \sum_{L=j+1}^{ORDP1} \sum_{a=1}^{j-1} X_r^+[ORD]\ B[L]\ \lambda_r^{L-(j+1)}\ \mathcal{Z}_\gamma[j-a]\ s^{a-1} \overset{!}{=} 0$$

$$\Lambda X^+ B = X^+ A$$

$$\tilde{\Lambda} \begin{bmatrix} X[1] \\ X[2] \\ \vdots \\ X[ORD] \end{bmatrix}^+ \begin{bmatrix} I & & & \\ & I & & \\ & & \ddots & \\ & & & B[ORDP1] \end{bmatrix} = \begin{bmatrix} X[1] \\ X[2] \\ \\ X[ORD] \end{bmatrix}^+ \begin{bmatrix} 0 & I & & \\ & 0 & I & \\ & & & \ddots \\ -B[1] & -B[2] & \cdots & -B[ORD] \end{bmatrix}$$

$$\tilde{\Lambda} X^+[1] = -X^+[ORD]\ B[1]$$

$$\tilde{\Lambda} X^+[2] = X^+[1] - X^+[ORD]\ B[2] \xrightarrow{\tilde{\Lambda} \times} \Rightarrow \tilde{\Lambda}^2 X^+[2] = \tilde{\Lambda} X[1] - \tilde{\Lambda} X^+[ORD]\ B[2]$$

$$\vdots$$

$$\tilde{\Lambda} X^+[L] = X^+[L-1] - X^+[ORD]\ B[L] \qquad \text{for } L = 2, \ldots, ORD-1$$

$$\xrightarrow{\tilde{\Lambda}^{L-1} \times} \Rightarrow \tilde{\Lambda}^{L} X^+[L] = \tilde{\Lambda}^{L-1} X^+[L-1] - \tilde{\Lambda}^{L-1} X^+[ORD]\ B[L]$$

$$= -\sum_{j=1}^{L} \tilde{\Lambda}^{j-1} X^+[ORD]\ B[j]$$

$a_1 = -X^+[ORD]\,B[1]$

$a_L = a_{L-1} - \Lambda^{L-1} X^+[ORD]\,B[L]$

$$\tilde{\Lambda}^L X^+[L] = -\sum_{j=1}^{L} \tilde{\Lambda}^{j-1} X^+[ORD]\ B[j] \qquad \text{for } L = 1, \ldots, ORD-1$$

$$* \quad \tilde{\Lambda} X^+[ORD]\ B[ORDP1] = X^+[ORD-1] - X^+[ORD]\ B[ORD] = 0$$

$$\xrightarrow{\tilde{\Lambda}^{ORD-1} \times}$$

$$\tilde{\Lambda}^{ORD} X^+[ORD]\ B[ORDP1] + \sum_{j=1}^{ORD-1} \tilde{\Lambda}^{j-1} X^+[ORD]\ B[j] - \tilde{\Lambda}^{ORD-1} X^+[ORD]\ B[ORD]$$

$$\Rightarrow \sum_{j=1}^{ORDP1} \underbrace{\tilde{\Lambda}^{j-1}}_{\text{diagonal}} X^{+}[ORD]\, B[j] = 0$$

$$\Rightarrow \boxed{X_r^{+}[ORD] \sum_{j=1}^{ORDP1} B[j]\, \tilde{\lambda}_r^{j-1} = 0}$$

e.g. $X_r^{+}[2]\,(K + \tilde{\lambda} C + \tilde{\lambda}^2 M) = 0$

in second order systems

$L = 1, \dots, ORDP1$ with $X_r^{+}[ORDP1] = 0$

$$\boxed{X_r^{+}[ORD] \sum_{j=1}^{L} B[j]\, \tilde{\lambda}_r^{j-1} = -\tilde{\lambda}_r^{L}\, \underbrace{X_r^{+}[L]}_{B[ORD] \text{ if } L=ORD}}$$

e.g. $X_r^{+}[2]\, K = -\tilde{\lambda}_r X_r^{+}[1]$ ✓

in second order systems

$X_r^{+}[2]\,(K + C\tilde{\lambda}_r) = -\tilde{\lambda}_r^2 X_r^{+}[2]$

$$\begin{cases} X^{+}[ORD] = X^{+}[ORD] \\ X^{+}[L-1] = \tilde{\Lambda}\, X^{+}[L] + X^{+}[ORD]\, B[L] \end{cases}$$

$B[ORDP1]$ if $L = ORD$

$X^{+}[ORD-1] = \tilde{\Lambda}\, X^{+}[ORD]\, B[ORDP1] + X^{+}[ORD]\, B[ORD]$

eg $X^{+}[1] = \tilde{\Lambda} X^{+}[2] M + X^{+}[2] C$

in second order systems

$X^{+}[ORD-2] = \tilde{\Lambda}\, X^{+}[ORD-1] + X^{+}[ORD]\, B[ORD-1]$

$= \tilde{\Lambda}^2 X^{+}[ORD]\, B[ORDP1] + \tilde{\Lambda}\, X^{+}[ORD]\, B[ORD] + X^{+}[ORD]\, B[ORD-1]$

$$\boxed{X^{+}[a] = \sum_{j=a+1}^{ORDP1} \tilde{\Lambda}^{j-(a+1)}\, X^{+}[ORD]\, B[j]}$$

$a = 1, \dots, ORD-1$

---

let $J_r[j] = \sum_{L=j+1}^{ORDP1} \lambda_r^{L-(j+1)}\, X_r^{+}[ORD]\, B[L]$

$$\begin{cases} J_r[ORD] = X^{+}[ORD]\, B[ORDP1] \\ J_r[j] = \lambda_r\, J_r[j+1] + X^{+}[ORD]\, B[j+1] \quad , \quad j = 1, \dots, ORD-1 \end{cases}$$

only $j \geq 1$ are used

let $Q_r[a] = \sum_{j=a+1}^{ORD} J_r[j]\, Y[1]\, \Lambda^{j-(a+1)}$

$$\begin{cases} Q_r[ORD-1] = J_r[ORD]\, Y[1] \\ Q_r[j] = Q_r[j+1]\, \Lambda + J_r[j+1]\, Y[1] \quad , \quad j = 1, \dots, ORD-2 \end{cases}$$

$$\left[\overbrace{J_r[1] + s\left(J_r[2] + s\left(\overbrace{J_r[3] + s\left(\overbrace{J_r[4] + \cdots}^{\text{ORD Terms}} \right)}^{\text{up To } J_r[\text{ORD}]} \right)\right)} \right] W_\gamma[1] \quad +$$

$$\left(Q_r[1] + s\left(Q_r[2] + s\left(Q[3] + \cdots \right)\right)\right) \begin{bmatrix} F_{\gamma,r} \\ F_\gamma^{(ext)} \end{bmatrix} \quad +$$

$$\left[\overbrace{\left(J_r[2] + s\left(J_r[3] + s\left(J_r[4] + \cdots \right)\right)\right)} \, z_\gamma[1] \quad + \right.$$

$$+ \left(\overbrace{J_r[3] + s\left(J_r[4] + \cdots \right)} \right) z_\gamma[2] \quad +$$

$$\left. + \left(\overbrace{J_r[4] + \cdots} \right) z_\gamma[3] \quad + \quad \cdots \right]$$